Lab 5: Viewing Life — Observing Structure and Function

BIOL-1

Name: ____________________ **Date:** ____________________

Learning Objectives

By the end of this lab, you will be able to:

1. Use a compound microscope to locate and focus on diverse biological specimens.
2. Make detailed macroscopic (visible to eye) and microscopic observations.
3. Infer the lifestyle and adaptations of an organism based on its structure.
4. Appreciate the diversity of life forms.

Overview

Biology is the study of life. In this lab, you will move beyond the textbook and observe living systems directly. You will be provided with three unknown biological samples: **Sample A**, **Sample B**, and **Sample C**.

These samples were collected by your classmates and brought into the lab, representing a "mystery box" of biodiversity. We do not know exactly what they are or where they came from. This mimics the real-world work of biologists who often encounter unknown organisms in the field and must use observation to understand them.

Your task is to observe them—first with your naked eye, then with a microscope—and deduce what they might be and how they live. This lab focuses on **careful observation**, **scientific inference**, and **connecting structure to function**.

Materials

- Compound light microscope
- Slides and coverslips
- Samples A, B, and C (brought in by class)
- Droppers/pipettes

- Lens paper

Part 1: Sample A

1. Macroscopic Observation (What does it look like to the naked eye?)

- Color, texture, movement, size, etc.

2. Microscopic Observation

Prepare a wet mount slide of Sample A. Scan at Low Power, then view at High Power.

Draw what you see at High Power.

Magnification used:

3. What is it? (Hypothesis/Identification)

4. How do you think it lives? What is it adapted to do?

(e.g., Is it photosynthetic? Does it swim? Is it a parasite? Support your answer with evidence from your observation.)

Part 2: Sample B

1. Macroscopic Observation (What does it look like to the naked eye?)

2. Microscopic Observation
Prepare a wet mount slide of Sample B.

Draw what you see at High Power.

Magnification used:

3. What is it? (Hypothesis/Identification)

4. How do you think it lives? What is it adapted to do?
(Look for structures like cell walls, flagella, cilia, or specialized shapes.)

Part 3: Sample C

1. Macroscopic Observation (What does it look like to the naked eye?)

2. Microscopic Observation

Prepare a wet mount slide of Sample C.

Draw what you see at High Power.

Magnification used:

3. What is it? (Hypothesis/Identification)

4. How do you think it lives? What is it adapted to do?

Conclusion

1. Compare the three samples. Which one was the most complex? Why?

2. What was the most surprising feature you observed today?